

Isoform-Aware Variant Annotation: Mapping Genetic Variants Across a Gene's Full Transcript Landscape

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2026 UCSF Summer Research Training Program

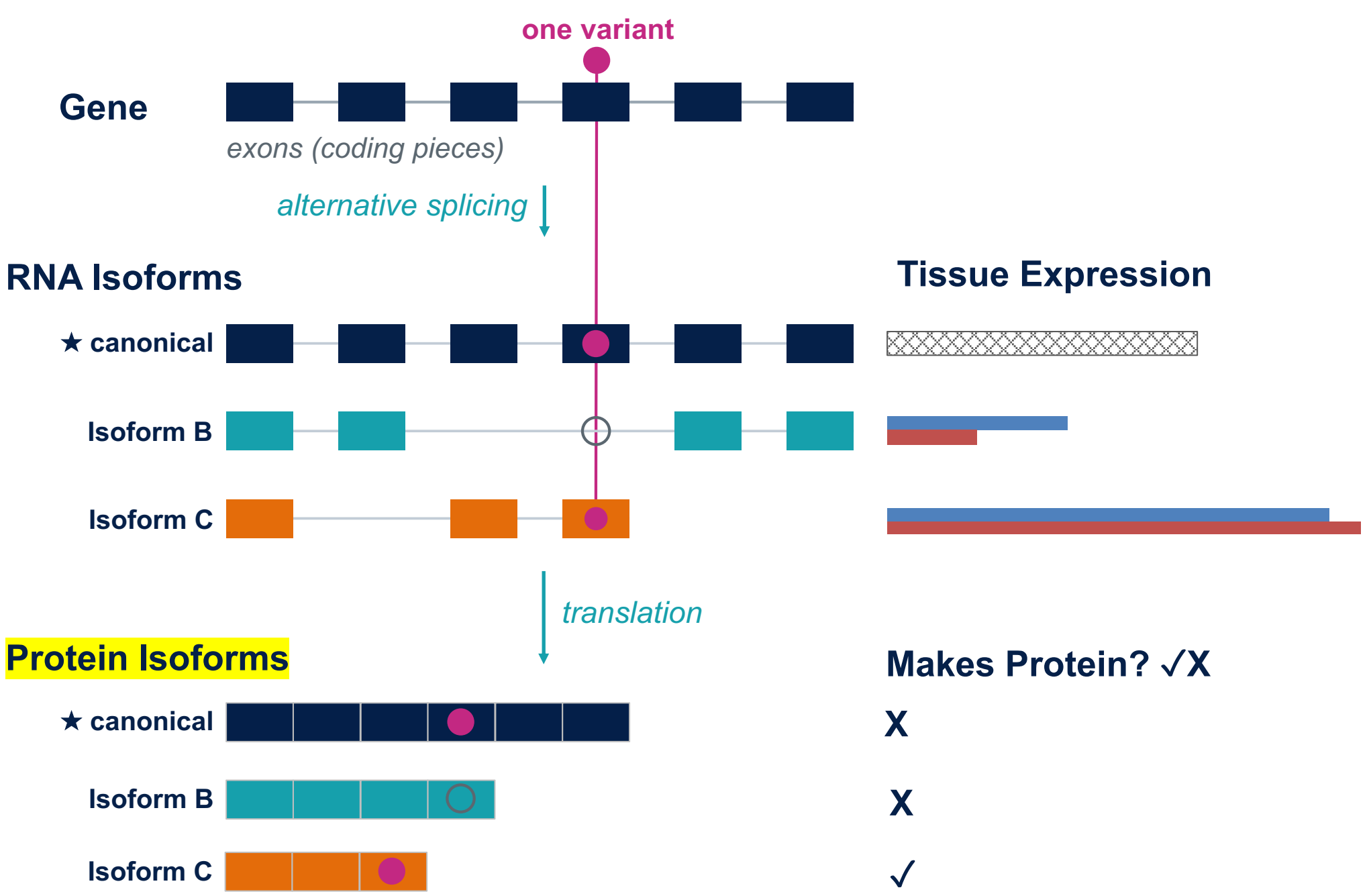


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Abstract

A single gene can produce many transcript isoforms, versions differing in which DNA segments are expressed. A genetic variant, a change in the DNA sequence, can have different consequences depending on the isoform: the same variant may be damaging in one isoform, yet absent in another. Standard annotation approaches ignore isoform-level diversity, considering a single "canonical" transcript, which may not be the most relevant in affected tissue. Reference isoforms, their expression levels in disease-relevant tissue, and associated protein domains are features needed to interpret variants in isoform context, but such information is spread across databases, on different genome builds and annotation versions, requiring substantial data harmonization. I built an isoform-aware annotation pipeline that aggregates and presents this data in a user-friendly format. The pipeline maps each variant across all of a gene's isoforms, covering both updated reference and novel brain-expressed isoforms from long-read sequencing, and reports, for every variant and isoform: the predicted effect, protein-domain context, and expression level in the human brain, highlighting how consequence and relevance vary across the isoform landscape. Because novel isoforms carry no annotated protein product, I extended the pipeline to predict each isoform's translation start site and the resulting protein, including its molecular weight, adding a layer no existing resource provides. We applied this to SYNGAP1, an autism risk gene, for which a 75 kDa protein is detected in neurons by western blot but for which no supporting reference isoform exists. The pipeline identified two brain-expressed isoforms that are predicted to produce a protein of this size and that patient variants map onto. This method generalizes to any gene or variant set, expanding our interpretation of variants and understanding of genetic etiology.

Introduction: One Gene, Many Isoforms



The same variant's effect differs

In a coding exon: Variant ALTERS the protein → But RNA not expressed

Exon spliced out: Variant NOT in this protein → And RNA IS expressed → But protein NOT made

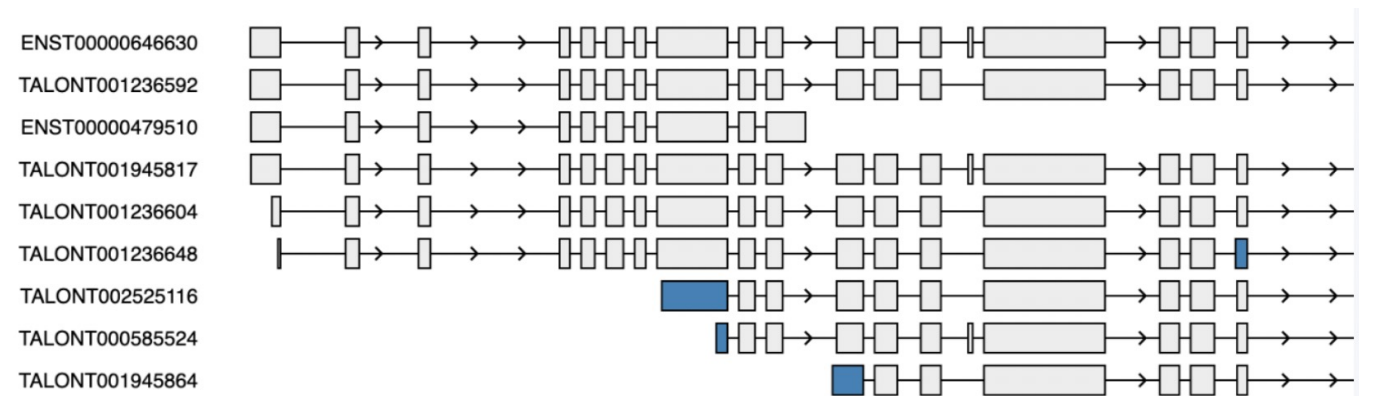
In a different exon: Variant ALTES the protein (differently) → And RNA IS highly expressed → And protein IS made

The Gap in Current Tools

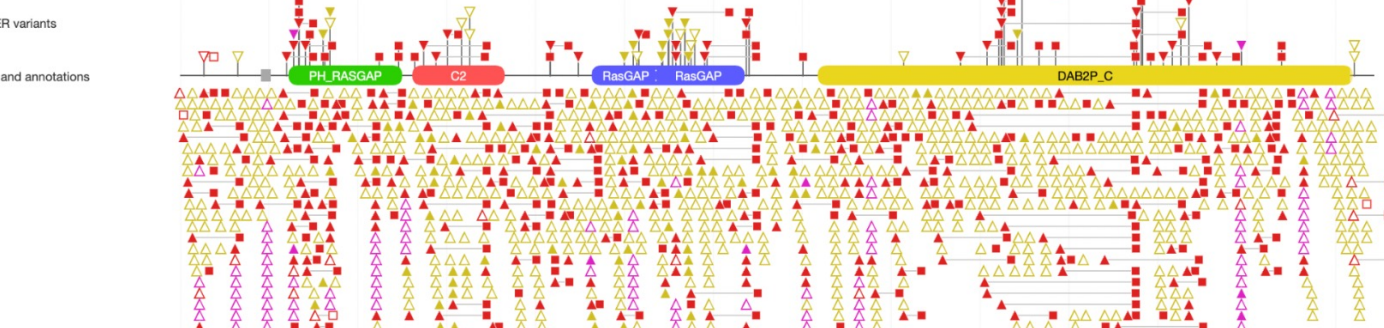
- See isoforms in detail (Ensembl): rich per-transcript annotation, but annotated "reference" forms only, misses novel isoforms

ENST00000646630.1	SYNGAP1-229	6015	1343aa	Protein coding
ENST00000629380.4	SYNGAP1-208	6026	1343aa	Protein coding
ENST00000644458.2	SYNGAP1-227	6025	1308aa	Protein coding
ENST00000449372.8	SYNGAP1-204	5977	1292aa	Protein coding

- See all isoforms (long-read isoform browsers): explore experimentally described novel isoform transcripts, but with limited detail and no variant or domain context.



- See all variants (DECIPHER): comprehensive patient variants, but mapped onto the canonical isoform alone.



Hypothesis and Objective

Hypothesis: Integrating patient variants with experimentally-described novel transcript isoforms can allow us to identify consequential isoforms beyond the canonical transcript, and, such as in the case of autism-associated gene *SYNGAP1*, protein domains of undescribed function.

- Objective:** build a gene-agnostic, isoform-aware annotation tool that places any variant list onto every isoform of any gene, in the context of protein domains, brain expression, and predicted protein products.

SYNGAP1 Case Study

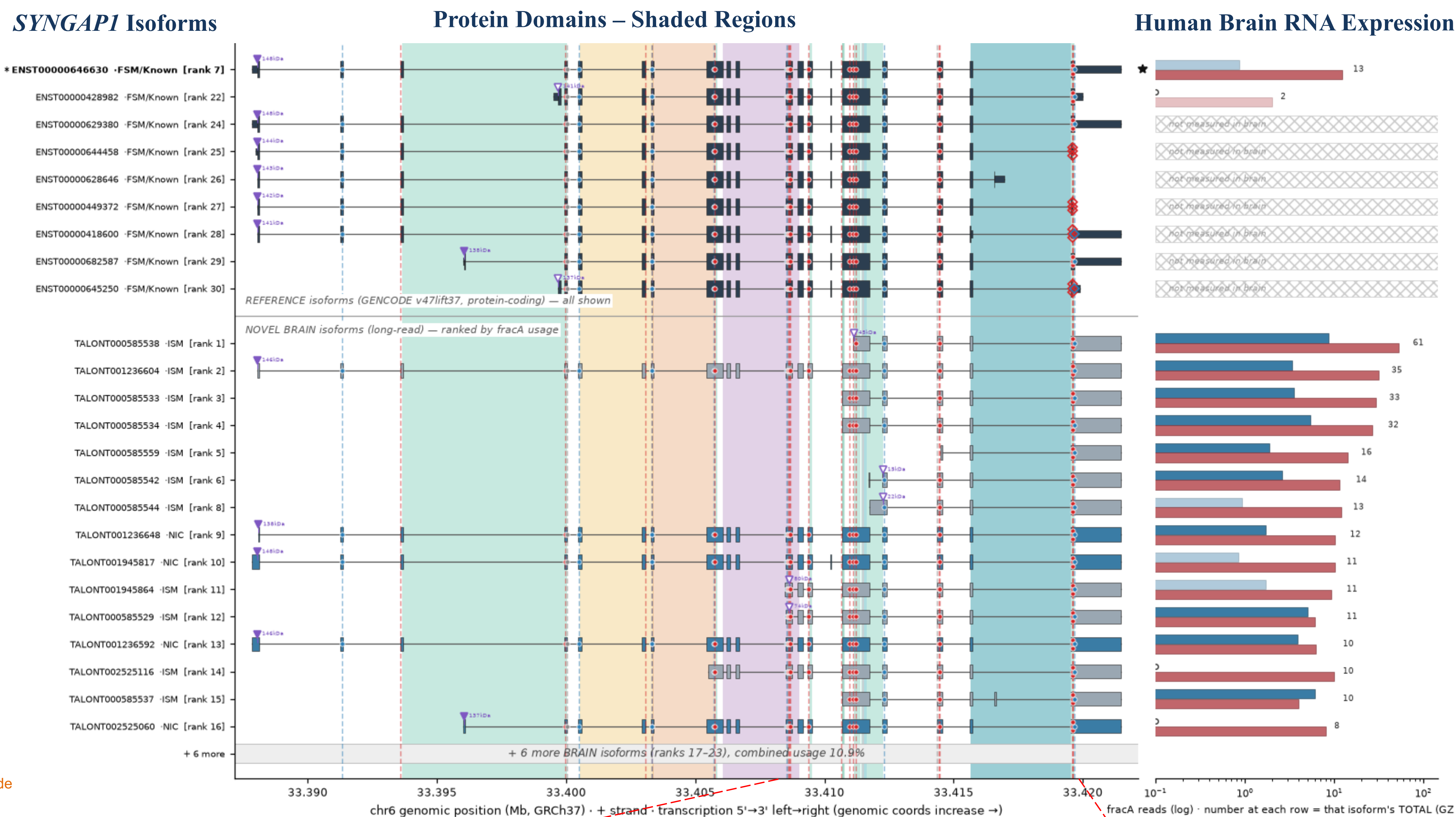


Figure 4a. Applied to *SYNGAP1*, the tool recovers brain transcripts that are absent from the reference annotation and that patient variants map onto: exactly the isoforms canonical-only tools ignore.

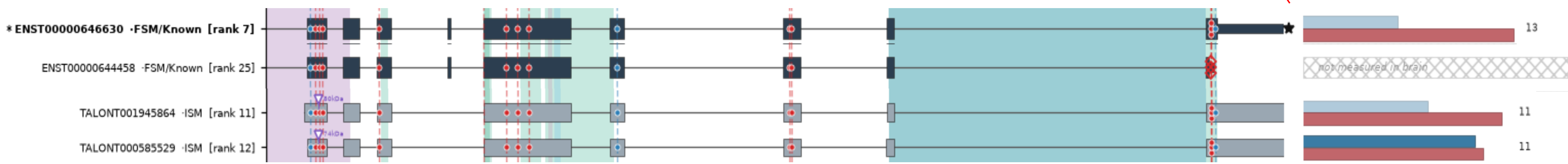
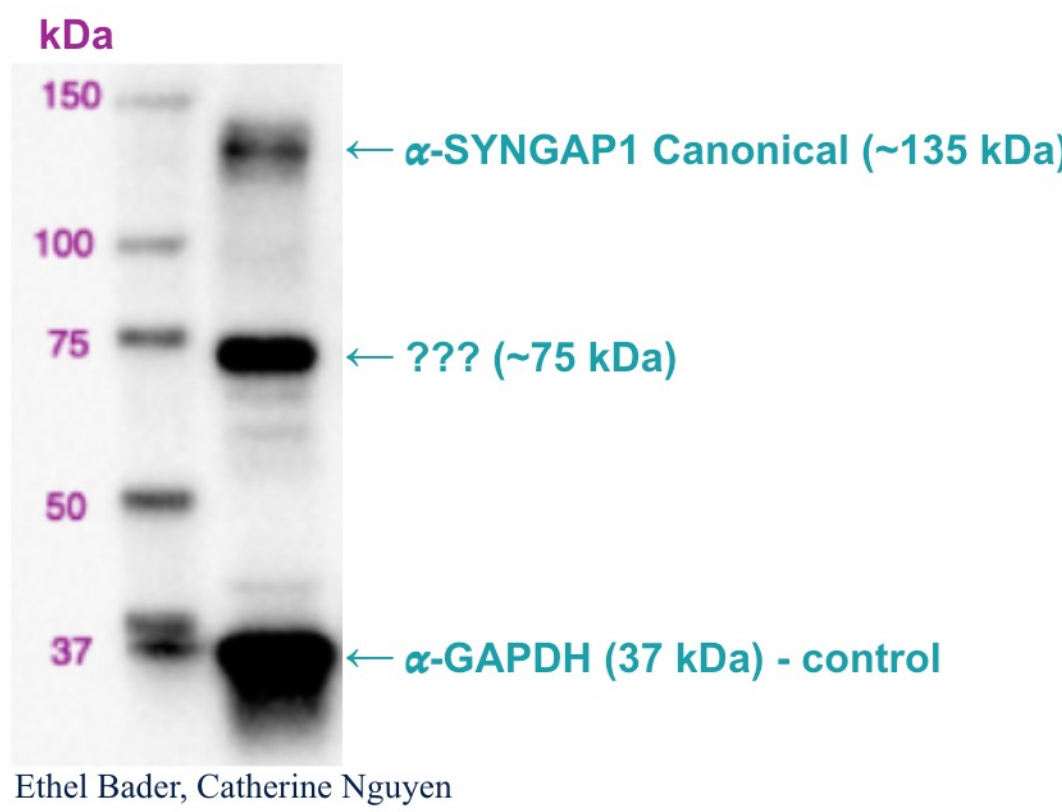


Figure 4b. Four isoforms, one variant set: the hypothesis in one view. All rows zoomed to the 5' start of the novel isoforms; both reference transcripts extend well upstream (not shown).

- Row 1 - canonical:** The transcript nearly all annotation is reported against. Read here, the 3' variants are truncating.
- Row 2 - a second reference isoform:** Those same variants fall in this isoform's 3'UTR, they're in the transcript, but not in the protein. However, the transcripts aren't seen in brain expression either way.
- Rows 3-4 - two novel brain isoforms:** Absent from the reference annotation, expressed in fetal cortex, and carrying patient variants. Both are short relative to the references and carry only part of the Ras-GAP domain; could give insight into novel domain functions.
- Why this matters:** an isoform that is brain-expressed, variant-bearing, and *incomplete in the canonical functional domain* is exactly the case the hypothesis predicts, a consequential isoform beyond the canonical, whose remaining sequence has no described function. Canonical-only annotation cannot represent it.

Linking Transcripts to an Unexplained Protein Band



Ethel Bader, Catherine Nguyen

Figure 5. Western blot of *SYNGAP1* in rat neurons. Alongside the canonical product near 135 kDa, the field had observed a band near 75 kDa that the known reference forms do not predict.

- The gap:** the tool described transcription, not translation, so it could not say which of these isoforms make protein.

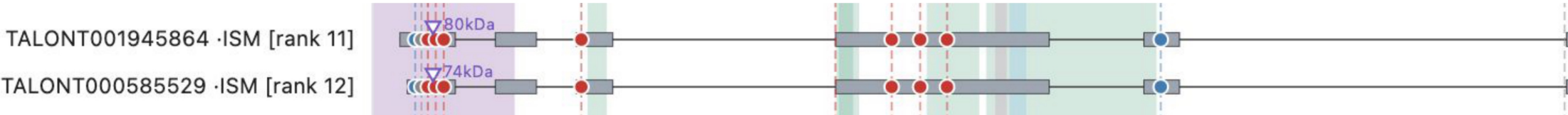
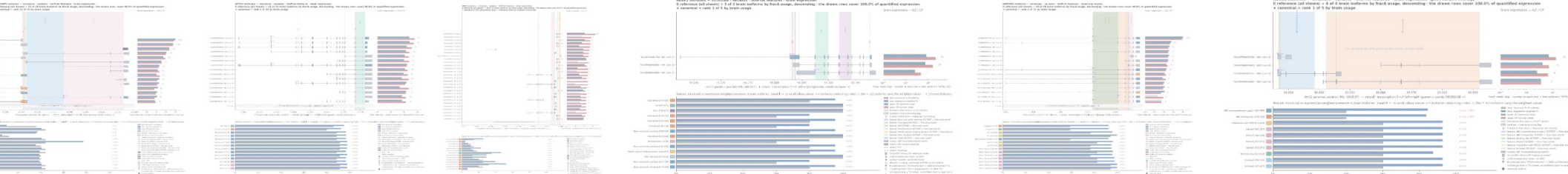


Figure 6. Two novel isoforms carry predicted translation start sites yielding products near 80 kDa and 74 kDa, in line with the unexplained band.

- Outcome:** adding predicted translation turned an unexplained band into two concrete, size-matched candidate isoforms. Wet-lab work is now underway to test whether either is the source.

Beyond SYNGAP1



SYNGAP1 became a deep case study, but the tool is GENERALIZEABLE.

For ANY gene + variant list, it will:

- Map every variant across every isoform
- Include experimentally-found brain forms
- Predict each isoform's protein product & size
- Read consequences in the context of domains

Conclusions & Future Directions

Conclusions:

- We addressed the need for a tool that integrates variant mapping with transcript isoform-level resolution
- Surfaced isoforms missing from the reference that patient variants map onto, and flagged two whose predicted products match a previously unexplained band
- The approach generalizes to any gene + variant list

Future Directions:

- Wet-lab validation of the SYNGAP1 protein
- Apply across more disease genes
- Integrate different tissue expression datasets
- Improve features & usability → public release

Acknowledgements

Thank you to my mentors Helen Willsey, Ph.D., Belinda Wang, M.D., Ph.D., and Matt N. Tran, Ph.D., and to the Willsey and Wang Labs. Supported by the UCSF Summer Research Training Program, UC LEADS, and UCSF Graduate Education and Postdoctoral Affairs.



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